

Elie ROUPHAEL, PhD

Postdoctoral Researcher in Control, System Identification and Learning

✉ elie.rouphael.98@gmail.com • Grenoble, France ➤ elierouphael.github.io

➤ linkedin.com/in/elie-rouphael ➤ github.com/ElieRouphael

Research profile

Postdoctoral researcher with a background in automatic control, mathematical modeling and data-driven analysis of dynamical systems. My doctoral work developed realization-theoretic and identification tools for stochastic switched and LPV state-space models, with emphasis on minimal representations, innovation forms and algorithms built from noisy data. My current postdoctoral work at GIPSA-lab, LIAS and Michelin concerns real-time estimation and model learning for vehicle dynamics and tire-road interaction. I am interested in rigorous methods that connect theory, algorithms and implementable estimation tools for complex dynamical systems.

Education

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| 2021–2025 | PhD in Automatic Control and Computer Science , Université de Lille, CRIStAL, France. Thesis: <i>Towards Stochastic Realization Theory for Linear Switched Models</i> . Advisors: Lotfi Belkoura and Mihály Petreczky. Defended 24 October 2025. |
| 2020–2021 | MSc in Automatic Control and Electrical Energy , Université de Poitiers, LIAS, France. Ranked first in class. Research project on non-stationary signal detection and frequency estimation. |
| 2016–2021 | Engineering Degree in Electrical and Industrial Control Systems , Lebanese University, Faculty of Engineering, Lebanon. |

Professional and research experience

Postdoctoral Researcher - LabCom I-TireLab

Nov 2025–present

GIPSA-lab (Grenoble INP–UGA), LIAS (Univ. Poitiers) and Michelin, Grenoble

- Real-time prediction of slippery road scenarios with hybrid modeling and machine learning, supervised by Guillaume Mercère and John-Jairo Martinez-Molina.
- Physics-informed and data-driven models for tire-road interaction and vehicle dynamics; Kalman filtering combined with data-based parameter estimation for real-time use.
- Digital-twin-based methods for tire-road interaction modeling and dynamics prediction.

PhD Researcher

2021–2025

CRIStAL, Université de Lille, France

- Stochastic realization results for generalized linear switched systems with inputs: deterministic and stochastic decompositions, innovation forms, minimality.
- Identification and realization algorithms for stochastic state-space models with switching signals, with numerical validation.
- Collaboration with SZTAKI, Budapest, on deep state-space and time-series models with generalization guarantees.

ATER and Teaching Assistant

2022–2025

Université de Lille, France

- ATER (2024–2025): 195+ hours of teaching in Python, C, control theory, microcontrollers and GRAFCET; supervision of 3 student internships and 1 bachelor's project.
- Teaching assistant (2022–2024): 128+ hours of tutorials and labs in automatic control, signal processing and feedback systems.

Research Intern

Mar–Jul 2021

LIAS, Université de Poitiers, France

- Detection of non-stationary harmonic disturbances in electrical systems (led to the IFAC 2023 paper below).

Publications

- ① E. Roupheal, M. Mejari, M. Petreczky, L. Belkoura, "Toward Stochastic Realization Theory for Generalized Linear Switched Systems With Inputs: Decomposition Into Stochastic and Deterministic Components and Existence and Uniqueness of Innovation Form," *IEEE Control Systems Letters*, 2024. DOI: [10.1109/LCSYS.2024.3417189](https://doi.org/10.1109/LCSYS.2024.3417189).
- ② E. Roupheal, M. Mejari, M. Petreczky, L. Belkoura, "Minimal Covariance Realization and System Identification Algorithm for a Class of Stochastic Linear Switched Systems with i.i.d. Switching," *IEEE Conference on Decision and Control*, 2024. DOI: [10.1109/CDC56724.2024.10886432](https://doi.org/10.1109/CDC56724.2024.10886432).
- ③ E. Roupheal, S. Cauet, A. Chamroo, "Variable Frequency Monitoring of Power Grid: A Modified Observer Approach," *IFAC-PapersOnLine*, 2023. DOI: [10.1016/j.ifacol.2023.10.1830](https://doi.org/10.1016/j.ifacol.2023.10.1830).
- ④ E. Roupheal, M. Petreczky, L. Belkoura, "On Minimal LPV State-Space Representations in Innovation Form: An Algebraic Characterization," *IEEE Conference on Decision and Control*, 2022. DOI: [10.1109/CDC51059.2022.9992964](https://doi.org/10.1109/CDC51059.2022.9992964).

Supervised student projects

- **PAC bounds for stable state-space models:** generalization bounds for deep state-space models, with SZTAKI (Budapest).
- **Classical vs. deep vs. state-space time-series models:** benchmark of ARIMA/ARMAX, RNNs, CNNs and Transformers against SSMs such as Mamba.
- **Transport-mode prediction:** predicting individual transport choices from Hauts-de-France data for mobility planning.
- **System identification toolbox:** open-source Python tools for LPV, hybrid and switched systems.

Technical skills

Programming	Python Matlab/Simulink C/C++ SQL R Git
Scientific Python	NumPy, SciPy, Pandas, Matplotlib, Scikit-learn; reproducible numerical experiments.
Machine learning	PyTorch, TensorFlow/Keras, JAX; neural networks, deep sequence models, physics-informed learning.
Systems and control	State-space modeling, stochastic systems, switched/LPV/hybrid models, observers, Kalman filtering, system identification, MPC.

Certificates and training

Universities	AI for Design and Optimization (University of Michigan, 2026); Bayesian Statistics: From Concept to Data Analysis (UC Santa Cruz, 2025); Neural Networks and Deep Learning (DeepLearning.AI, 2023).
IBM	Deep Learning and Reinforcement Learning ; Specialized Models: Time Series and Survival Analysis ; Deep Learning with Keras and TensorFlow ; Introduction to Deep Learning & Neural Networks with Keras ; Machine Learning with Python ; Generative AI and LLMs: Architecture and Data Preparation ; Python for Data Science, AI & Development ; Introduction to Data Engineering ; Python Project for Data Engineering ; Introduction to Relational Databases ; Introduction to DevOps .
Schools	EECI course on Sparsity and Big Data in Control and ML; spring school on closed-loop and nonlinear system identification and deep learning; ACES summer school in modern control theory (60 h).
Languages	Arabic - native French - B2, DELF English - TOEIC 865/990

References

Prof. Lotfi Belkoura
Professor, CRISAL, Univ. Lille
lotfi.belkoura@univ-lille.fr

Dr. Mihály Petreczky
CNRS Researcher, CRISAL, Centrale Lille
mihaly.petreczky@centralelille.fr

Prof. Guillaume Mercère
Professor, LIAS, Univ. Poitiers
guillaume.mercere@univ-poitiers.fr

Prof. John-Jairo Martinez-Molina
Professor, GIPSA-lab, Grenoble INP
john-jairo.martinez-molina@grenoble-inp.fr